



GASLAB NEXUS

-Honors Project Year 1-

Smart Cities Track

Coach: Floor van Schie

Students:

Maria Barnechea Dominguez

Andreea Vasile

Valentina Wu

Marta Zawadka

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1. | The team

Throughout the year the team dynamics have altered and adjusted naturally both because of gaining experience, confidence and expertise within the project as individuals and as a team; and as the tasks to be performed changed when the project evolved and became more technical and specific. As a result of our backgrounds and different areas of interest, as well as our individual personal development goals, we each fell quite seamlessly into the different roles within the team. However, even though each of us was focusing on different project areas the collaboration and communication within the team stayed consistent and we have supported each other in this first year of Honors Academy. Important decisions however, such as design plans, the overall direction of the project and many more were still taken as a team and input from everyone was appreciated and taken into consideration.



My name is Maria, I am a second-year student in the faculty of industrial engineering, where I am focusing on the fields of operations management and financial models. I joined the Gaslab project as an academic and personal challenge where I would have the opportunity to exploit my creativity and delve into the new field of architecture and built environment.

My name is Andreea, and I study Architecture and Urban Design. I enjoy working in collaborative, interdisciplinary settings and am passionate about functional, sustainable and user-centered design. This project is a great opportunity for translating knowledge and research into spatial strategies that enhance user experience and contribute to long-term adaptive and responsible design.



My name is Marta, and I am a student in the field of Architecture and Urban Design. My studies are primarily focused on sustainability, architectural design, and understanding the needs and experiences of future stakeholders. This project is an opportunity for me to bring together my creativity, problem-solving skills, and passion for designing thoughtful, future-oriented spaces.

My name is Valentina, and I am following the Building Physics and Services track within the Built Environment. I am passionate about user-centered design strategies, especially for its comfort and indoor environmental quality declination. I believe this project to be a great opportunity to apply my technical knowledge to a real building renovation case.



2. | Project description

The aim of this honors project is to adaptively reuse the historical Gaslab building. The look and feel of the former high-pressure chemistry laboratory will be preserved and integrated with new functions that meet the future needs of the university. Our 'client' or the most significant stakeholder is Innovation Space, which aims to have an "**experience center**" and provide an inspiring and flexible space for interdisciplinary collaboration.

Currently, the building is outdated and used without a defined function, as student teams are temporarily housed and there are no clear future plans that have been established. Moreover, the interior and Gaslab's structure no longer meet contemporary standards, as the building has not undergone any significant renovations since 2002. Additionally, the historical significance of Gaslab is not emphasized through its function and usage. Therefore, the building, which is now not connected to other educational facilities, has lost its 'visibility' and popularity among the university's community.

Our goal is to conduct a thorough analysis in order to develop a comprehensive understanding of the building and develop a design that responds to the current problems and needs of both students and working professionals. In this way, the renovated Gaslab would be transformed into a multifunctional space — flexible and easily adaptable to changes over time — while maintaining a clear and fixed purpose to ensure optimal functionality. Additionally, the building will regain its significance and will be displayed as a nexus for inspiration and collaboration, accessible for both the university's community and external visitors.

Therefore, the **research question** that leads our project is:

What is a sustainable and functional new purpose for the Gaslab building on the TU/e campus?

Throughout the year, in-depth research and analysis regarding the urban context, as well as the indoor environmental quality of the building, was conducted. Together with active involvement with relevant stakeholders and confrontation with TU/e Masterplan vision, Gaslab, originally a chemistry laboratory, will be repurposed as an adaptable, flexible and future-proof space combining:

- A **depot** and **exhibition area** for the TU/e art collection;
- **Collaborative student spaces** for meetings, studying and interdisciplinary events;
- And a **publicly accessible venue** for academic ceremonies, events, talks and creative engagement.

3. | Methodology and chosen approaches

3.1 | Stakeholder mapping and outreach

The project's most time-consuming phase was identifying and aligning with the correct stakeholders. The process started at the beginning of the Honors year, as a first and crucial step to brainstorm viable options for a new function for Gaslab, as well as possible limitations posed by real estate and building constraints. This step took several months due to unclear management positions for the building and conflicting future renovation plans that involve the campus on a broader scale.

The project's direction changed at multiple instances throughout the year and only settled in recent months, after starting an active correspondence with the Gaslab managerial department from Innovation Space.

Despite the extensive time consumed on this aspect of the project, this iterative process allowed the creation of a comprehensive stakeholder map and feedback matrix. Our team gained a better understanding of Gaslab's evolution and usage patterns through the lens of current and former users, contributing to the development of a design plan that centers users and their experiences. Furthermore, stakeholder visions were consolidated to guide design constraints, function feasibility, and compliance with TU/e's strategic goals.

This section provides an overview of the main stakeholders that have been contacted throughout the project so far.

The first category of stakeholders that we reached out to were **past and current users** of Gaslab. Knowing that the building changed its function multiple times before being transformed in a co-working space that now houses student teams, we investigated the past stages of the building and contacted **EAISI** (Eindhoven Artificial Intelligence Systems Institute), **Stodium Generale** and **student teams** (VOID, Polar, HART, Serpentine, Fruitpunch AI), which all shared feedback on past usage patterns, as well as potential strengths and weaknesses of the building's comfort and functionality. This output generated an overview of the main issues that we aim to tackle such as roof problems, poor indoor environmental quality and lack of flexibility and storage.

TU/e Real Estate is responsible for the entire real estate development of campus. It caters for the project management and management of all buildings, installations and infrastructure. For this project to be approved and developed, it is crucial for this entity to approve all development plans and budgets.

The **Green Office** was also important in guiding us through sustainable construction practices, and giving us access to useful documents that provided valuable insight into the desired building characteristics and goal of the university. They also provided insight into the recent renovation of Atlas and its recently acquired BREEAM certificate, and whether or not achieving a similar honorary for Gaslab was feasible given its characteristics.

The **Quality Committee** is in charge of reviewing architectural, urban design, and landscape plans, making sure that all new buildings and renovations align with the 2040 Master Plan and landscape vision. The master plan proposes a set of urban and architectural principles and phasing of new constructions to adapt the campus to the contemporary needs and requirements, maintaining a homogenous look and feel throughout the TU/e grounds, thus providing more insight on heritage considerations, building alterations, and potential design constraints. Nevertheless, the meeting with one of the representatives of the committee clarified our uncertainties and generated a better understanding of the various, sometimes strict, policies and frameworks for the buildings on campus redevelopment. It was a crucial moment for receiving approval for innovative and creative building functions, that are not only encouraged for the repurpose of the building, but are also perfectly aligned with our ambitions and visions.

Innovation Space provides students at TU/e with an opportunity to engage in challenge-based learning and develop and prototype new ideas. At the moment, it is the organism in charge of the management of Gaslab. Therefore, our team is closely collaborating with its members to create a space where TU/e students, partners and institutes can continue to create, test and develop new technologies and ideas. Via multiple meetings with the managerial department from Innovation Space responsible for Gaslab, their initial “Experience Center” concept, as well as ongoing involvement and future plans were revealed and taken into consideration for our design proposal.

TU/e Art Collection Committee provided relevant input on the possibility of displaying the extensive art collection that TU/e has the privilege of owning, which showcases the history and heritage of the university. The committee also manages the expansion, maintenance and accessibility of the TU/e Art and Heritage Collections. The aim of this committee is to keep both the material and immaterial collections relevant for future generations. Our project thus envisions a flexible, adaptable space that functions both as an exhibition venue for the pieces of this collection, as well as a hub for inspiration, dialogue, and idea-sharing. In our design, the building itself becomes part of the exhibition, with its architectural remains narrating its own story while reflecting the broader history of the campus. By blending technology and art, we aim to maximize the potential of the building’s space and highlight its cultural significance.

The latest collaboration that we initiated is with the **Digital Twin Lab**, which is a part of the EAISI research lab that deals with modelling real-world situations and creating data interfaces between the real context and model by using tools such as virtual reality and augmented reality. With their help we hope to bring the project into a new realm of possibilities, including creating new and exciting virtual exhibits; bringing art to life; allowing for a more interactive and in-depth experience exploring the history of Gaslab and the TU/e campus; creating a more interactive and learning environment; creating Gaslab renders and prototype to create a tangible visualization of Gaslab and work on the renovations in real time.

3.2 | Spatial and urban context analysis

The urban analysis constituted a significant step for understanding the contextual surroundings and spatial dynamics of Gaslab. Conducted at the scale of the entire campus, this analysis helped us identify **spatial patterns and hierarchies** between the various buildings. Moreover, important information was extracted from both observational and data planning documents, including *TUe Masterplan 2040* and *Landscape Vision TU/e Campus 2040*. These documents enabled us to make a connection between the team's analysis and strategic spatial structures that shape the broader **campus morphology** and **Gaslab's contextual placement**.

The two foundational frameworks provided by the TU/e staff articulate a cohesive vision for the university campus that is rooted in modernist planning principles and emphasizes an open, omnidirectional campus integrated into the urban fabric of Eindhoven. These extracted statements, along with the urban analysis generated one of the main principles used throughout the entire design process: **connectivity**. Therefore, from the key positioning of Gaslab, on the southern edge of the campus, the Gaslab building can become a link between Eindhoven city center and the university's community.

Focus areas:

- **Accessibility & Location:** Mapping of entrances and main paths on campus, green zones and building orientation.
- **Architectural Identity:** Gaslab is a freestanding, double-height structure with four equally important facades. It is not listed as a monument, but heritage value must be respected.
- **Surroundings:** Landscaping and outdoor usage (e.g. urban greenery, parking transformation) were considered to enhance the public-facing nature of the building.

Outcomes:

The drawings for the analysis of maps had a significant role in identifying urban challenges and opportunities that could later be incorporated in the design considerations for the repurposing of the building. Therefore, the following section describes three areas of study and states their conclusions and relevance for the latter approaches and design considerations.

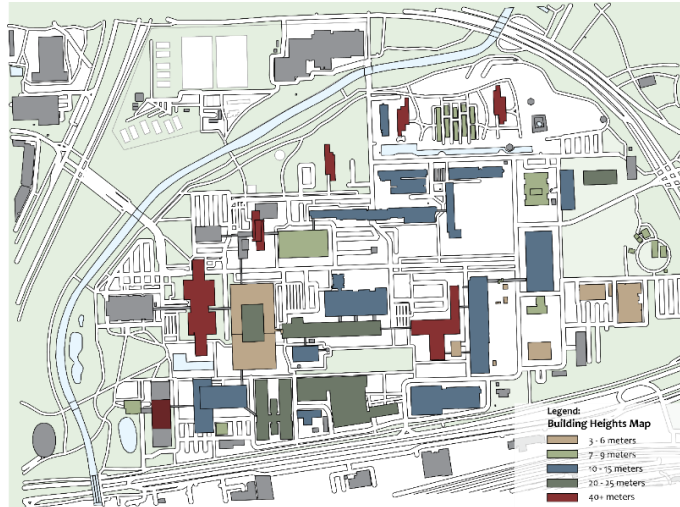


Figure 1

Positioned next to Vertigo – one of the architectural landmarks of the campus – and Matrix, Gaslab is surrounded by taller buildings that block the view from the center of the campus towards it (Figure 1). This was one of the earliest identified challenges that gave us ambition to transform this old building into an attractive center through innovation. Therefore, a new attractive function and spatial layout will spark the interest of the public and regain Gaslab’s popularity and ‘visibility’.

Additionally, the urban and spatial analysis informed strategic considerations for integrating solar energy into the transformation of Gaslab into a Living Lab fully powered by renewable sources. The study revealed that installing solar panels on Gaslab's roof would be inefficient due to persistent shading caused by the surrounding, taller buildings. As a result, alternative strategies need to be explored, such as relocating solar installations to other, more sun-exposed areas of the campus or implementing other sustainable and efficient systems.

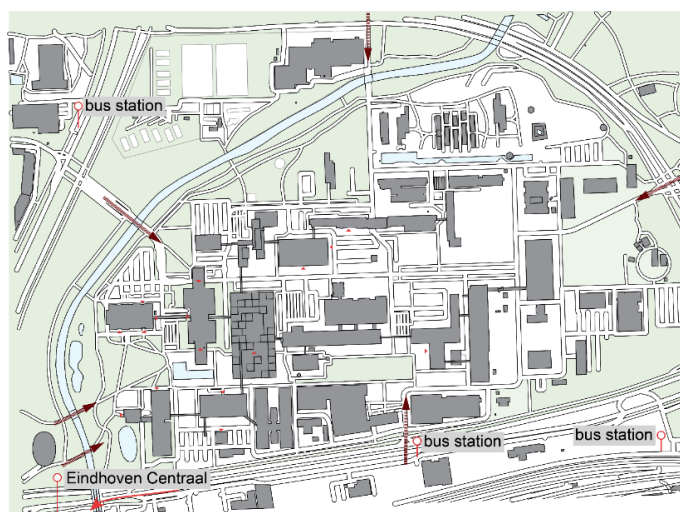


Figure 2

The Mobility and Accessibility analysis showcased the available routes for accessing Gaslab (Figure 2). Moreover, this study confirmed the advantageous location within campus for both students or staff and other external visitors. Additionally, the area is now easily accessible by all modes of transportation (cars, bikes and pedestrians), including covered bike parking and above- ground car parking. The car dominance in the surroundings of Gaslab represents a threat to its attractiveness, but the *Landscape Vision TU/e Campus 2040* outlines the ambition of reinforcing the ecological and recreational character of this part of the campus. Thus, our project has the opportunity to introduce greenery into the surroundings.

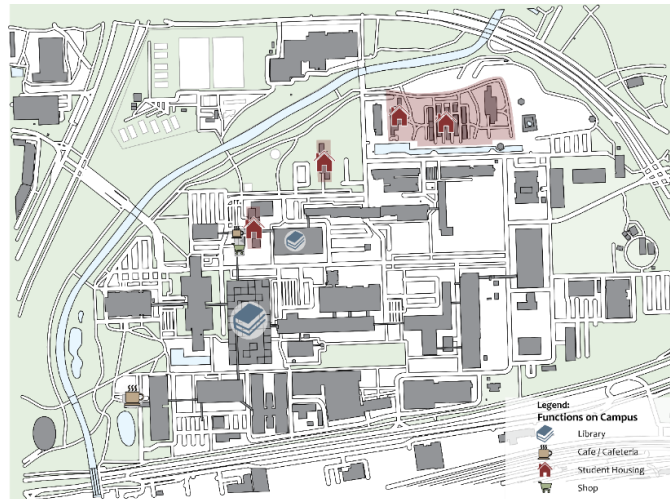


Figure 3

The functional analysis of the surrounding buildings provided valuable insights into their usage patterns (Figure 3). Gaslab is positioned behind Matrix - an innovation hub - and nestled between educational buildings from various faculties. This location presents a unique opportunity to transform Gaslab into an inspiring focal point for campus life.

3.3 | Building diagnostics & physical assessment

The scope of the project included maximizing the spatial flexibility to allow adaptive usage of the building: this led to multiple qualitative inspections of the existing structural layout, to evaluate the possibility of installing movable partitions and display elements.

For the assessment of the indoor environmental quality of Gaslab, on-site measurements and evaluations were conducted, specifically targeting **acoustics, lighting, and temperature**. Upgrading these aspects improves **user's health and comfort**, as well as generally contributing to the alignment with sustainability goals by reducing energy consumption and preserving the environment.

For the measurements, collaboration with the Building Physics Laboratory was initiated to access the necessary equipment, which included hand-held tools, such as the Illuminance meter T-10A and the Testo 605i.

An overview of the steps and intended outcomes for the building physics section is provided below.

Lighting

Step: Compile measurements regarding artificial lighting levels, with particular focus on working areas and surfaces (Figure 4).

Outcome: Find a suitable solution for the different activities taking place in Gaslab (i.e. prototype building, meeting, presentations, etc.).

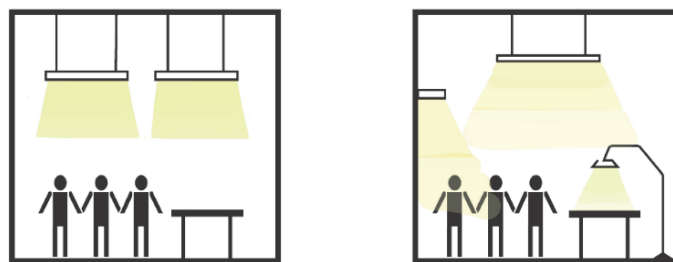


Figure 4

Heating

Step: Measure indoor temperature in critical areas – where people usually work – as well as zones regarded as problematic based on interview with the building's current past and current users, as common feedback was the uneven temperature distribution.

Outcome: Improve heat distribution and reduce vertical temperature gradient, to efficiently regulate temperature of the building and minimize energy loss (Figure 5).

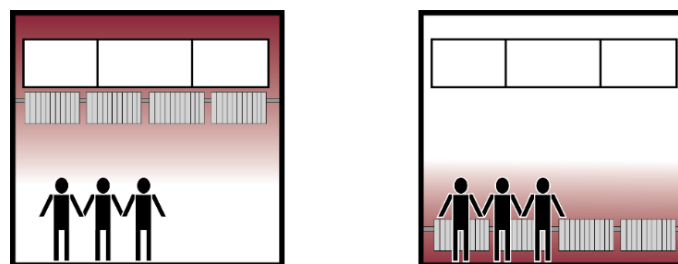


Figure 5

Insulation

Step: Conduct qualitative analysis of current thermal insulation layer, using a roof section retrieved from recent construction work carried out in Gaslab for the installation of a service staircase.

Outcome: Improve air tightness and reduce thermal bridges and heat loss, while preserving the extensive glazed area, which is a distinctive characteristic of Gaslab (Figure 6).

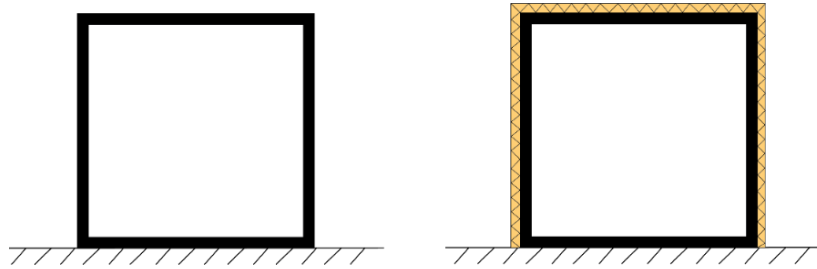


Figure 6

3.3.1| Program of Requirements

As an integral part of the project, a Program of Requirements was developed to outline standard conditions for the **indoor environmental quality** of Gaslab. The boundaries are primarily based on recognized frameworks, such as the Frisse Scholen 2021 and the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE).

To determine the current quality of the building, measurements will be carried out and findings will be compared with the Program of Requirements: this will be the basis of suggestions for building comfort improvements, tackling lighting, thermal comfort, as well as refurbishing the building envelope, for instance by improving insulation and airtightness.

Lighting

Regulations regarding lighting conditions are set in the Frisse Scholen regarding the illuminance by artificial light at work surface level, which should be at least 500 lux.

Thermal comfort

The combination of the air temperature and radiation temperature should be maintained between 18 and 25°C in the heating season. In this report, the indoor temperature will be set at 21°C and the calculations will be computed considering the average outdoor temperature of 5°C.

A sliding temperature scale applies to the temperature in summer and the intermediate season, where the temperature limits inside increase slightly with the outside temperature according to the following formula:

$$\text{Operative temperature (indoor)} = 0.33 \times \text{current average outside temperature} + 16.4 \pm 4^{\circ}\text{C}$$

The measurements were conducted in March, when the outside temperature was approx. 12 °C, thus resulting in a 24.4 °C upper bound threshold.

Acoustics

The airborne sound insulation between a working area and a (quiet) learning area is at least 31 dB.

3.3.2 | Measurements

A quantitative analysis was performed by conducting on-site measurements. This took place at daytime to account for the daylight availability, as well as the acoustic nuisance from the presence of working student teams in the building and external noise sources. As for the temperature, the findings might be limited because the measurements were conducted in March, thus they do not fully reflect the thermal discomfort that might be experienced during periods of more extreme weather (excessive cold in winter and overheating in summer).

Temperature

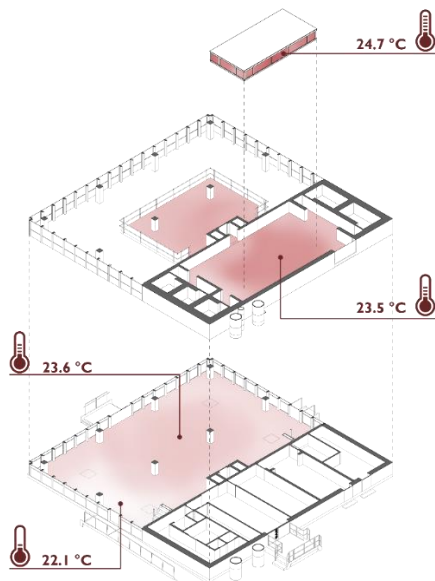


Figure 7

The measured temperature has non-uniform distribution between ground floor and first floor, and also across different areas on the same level (Figure 7). This is a sign of the inefficiency of the current heating system and inadequacy of the current thermal regulation.

As Gaslab features an open floor plan, the position of heating and cooling devices significantly influences the uniformity of the temperature.

Comparing the measurements with the Program of Requirements, the only location that surpasses the upper threshold of 24.4 °C is the top glazed layer. Despite falling within the guidelines, direct interviews revealed that the temperature is not perceived as totally comfortable by the buildings' users.

Acoustics

The reference value shown in Figure 8 is the LA90 extracted from the acoustic measurements, which indicates the percentile sound pressure level exceeded for 90% of the measurement period.

Comparing the findings with the threshold value of 31 dB set in the Program of Requirements, it can be deduced that the acoustic environment needs significant improvement, to enhance both user comfort and concentration in the working or meeting areas. One aspect that can be noted by visiting the building on a busy day is the high level of noise contamination from different areas of the building: by standing at one point, sounds from across the floor area can be heard and it is quite disturbing and distracting.

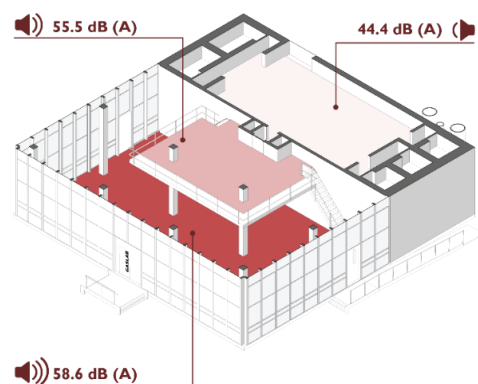


Figure 8

Lighting

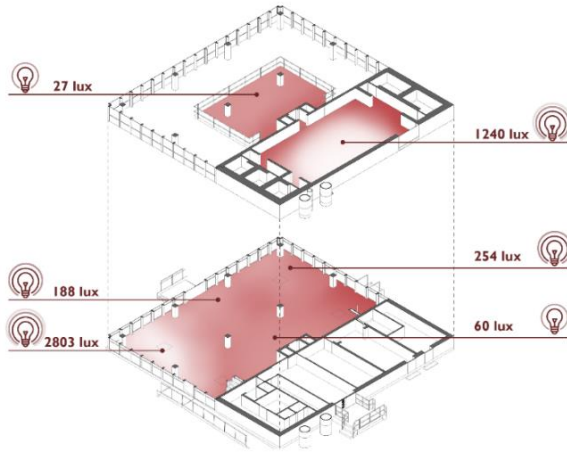


Figure 9

The lighting measurements target the quality of artificial lighting in the building, particularly the illuminance values were used as the assessment parameter.

As seen in Figure 9, the lighting uniformity is not reached across the floor area: perimetral zones register significantly higher values due to proximity to the extensive glazed areas that characterize Gaslab's envelope. However, more internal areas do not meet the 500-lux requirement, even with artificial lighting. For some areas, which are not supposed to function as working spaces, the requirement is not as strict, but generally, lighting needs improvement to allow adequate visibility for working surfaces as well as fostering a more pleasant atmosphere.

3.4 | Determining the historical use of the building

To properly assess and articulate potential uses and projects for Gaslab it was crucial to develop an understanding of the history of the building (Figure 10). Gaslab was originally completed as a part of the Chemistry ensemble on the TU/e campus in 1960. Since then, it has served multiple functions and purposes but always maintaining its **distinctive industrial character**, being the last building of the chemistry ensemble that maintains its original appearance.

It was first built between 1957-1965 during the first construction period undergone on the TU/e campus. In this phase the main focus of the architect was to create a unified design approach to create buildings with a clear functionality and a **cohesive look and feel**. In the building the functional modernism design philosophy employed by Van Embden can still be appreciated in modern times. The intended function of Gaslab in this period was to be a high-pressure laboratory, to host experimental and hazardous procedures. The robust brick, open section and industrial elements that are still present in the building act as testimony of this era.

During the 60's and 70's expansion plans for Gaslab were proposed to transform the building into a central safe laboratory, these plans, however, never came to fruition. Nevertheless, modifications were made within the building to address overcrowding and safety concerns.

By the late 20th century, due to a change in needs on the part of the chemistry faculty, many of the original buildings were demolished and Gaslab was renovated in 2002 to act as a venue for cultural activities for Studium Generale. This departure from its original function meant the addition of a new floor and the removal of some of the experimental instruments. Other issues such as increasing the accessibility of the building and adding sanitary facilities were also addressed.

Finally in 2011, the main entrance was shifted and, following more demolitions, Gaslab became more unique on campus as the sole remaining building of the original chemistry ensemble. Today, Gaslab is not only valuable as a useful piece of real estate, but as a **testament to the history of the university**, acting as a time capsule for the past.

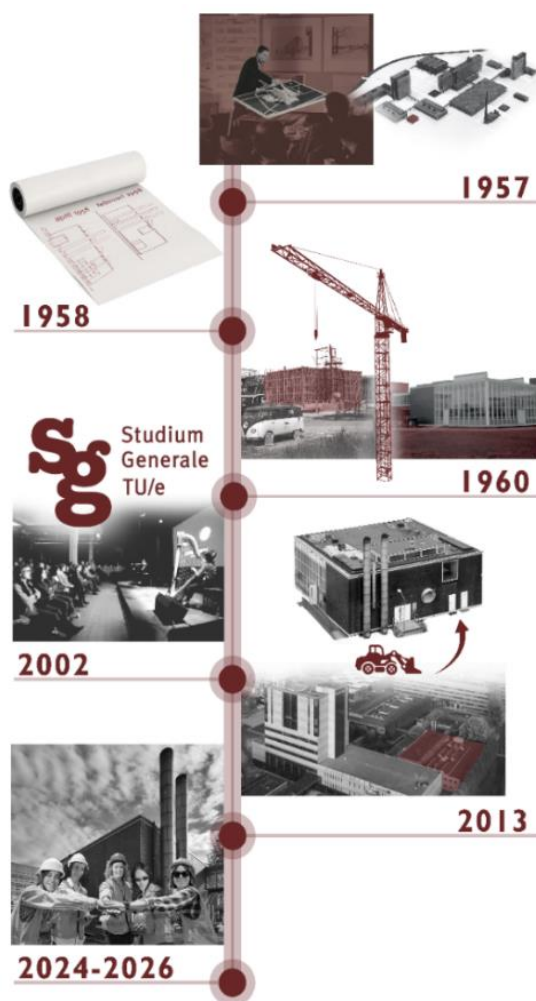


Figure 10

4. | Challenges faced

4.1 | Time

Several weeks were spent contacting various departments across the TU/e network, chasing replies, and clarifying responsibilities – for instance, who owns or manages the building – resulting in an extensive amount of time lost in stakeholder identification, which caused the project to suffer multiple setbacks, thus slowing down the active design process.

Despite these difficulties, this step was essential, as one of the project's goals was to respond to stakeholder needs and visions. Engaging and finding the right people was critical for ensuring relevance and feasibility, and although time-consuming, **stakeholder involvement and participation** could not be overlooked or disregarded.

4.2 | Documentation gaps

As one of the requirements set for the project is the **preservation of the historical character and value** of the buildings' components, a thorough analysis of the original plans and past proposals for renovation needed to be conducted. However, these resources were not openly published and accessible, causing difficulty retrieving construction details and creating an initial documentation gap. Building regulations were also not well defined, thus making it unclear within which boundaries we could explore and develop design concepts.

One example of this challenge regards the fire safety staircase that connects the first floor to the ground floor. Originally, the building had two symmetrical staircases, but for fire safety regulations, one of them was eventually closed off. As one of our design ideas, we wanted to restore the original symmetry by uncovering the now-hidden staircase. We tried to contact the fire safety management to receive consultation on the feasibility of this option, as explicit regulations and limitations were not available, but that was unsuccessful, and after meeting with the managerial department of Gaslab, the constraints remained unclear. However, a potential solution that combines aesthetic and safety purposes was eventually found and implemented in our final design proposal.

4.3 | Conflict of interest

The challenge consisted in finding an intersection between stakeholders' plans and our own vision for the project. The direction of the project and the proposed design had to align with existing plans, regarding future campus development, as well as smaller scale vision specifically for Gaslab, outlined by Innovation Space. These boundary conditions required **diplomacy** and **design negotiation** with both the stakeholders currently involved and the external parties responsible for the building management, such as the TU/e Real Estate. Only after multiple setbacks and detours from the initial concept brainstorming, a compromise was reached, which finally allowed the project to get a foothold, thus initiating the detailing and prototyping phase in preparation for the Final Demo. It is important for us to highlight that our current network of collaboration comprises of parties open to sharing their needs, while also valuing when we **take our own stand as a team**.

5. | Key decisions and rationale

The main drivers behind the selection of the proposed function and design are the need for informal collaboration and public engagement spaces on campus, the lack of **visibility** of TU/e's extensive cultural heritage, as well as the pre-existing plans for enhancing **flexibility** and innovation capacity of Gaslab.

Therefore, our proposed solution for the building's new function is an **art depot and exhibition** space, that showcases the historical and cultural significance of TU/e, through archival pieces from the Art Collection. This will be combined with adaptable spaces for collaborative student use (i.e. study, meetings, academic events, contact with external companies and institutes, etc.), thanks to the modular and movable arrangement of the art display surfaces.

Moreover, our goal is to launch Gaslab as an innovation **hub** that brings together **past legacy and future excellence**, by implementing new features, such as a VR interactive tour of the campus and history of Gaslab.

Overall, although the building is not a protected monument, changes must respect the architectural identity of Gaslab (e.g., freestanding visibility, curtain wall). The proposed implementations are modular and prefabricated to limit the invasiveness of the construction procedure, and no alterations to the original structure and components are planned to be carried out.

6. | Output and deliverables

The following section compiles all the deliverables produced over the course of this academic year. They showcase the effort, passion and time invested in the process towards the success of the project, as well as providing an insight into the thought process behind the design choices that led to the final project proposal.

6.1 | Midterm Demo

6.1.1 | Introduction

For the Midterm Demo, our vision was to create a **multifunctional space** that includes housing for student teams and spaces suitable for meetings and different types of events and activities organized by Innovation Space or by university's Institutes. Our main objective was to introduce different design strategies for **flexible and adaptable spaces** that respond to **diverse user needs**. These ideas were initially shaped through discussions with Innovation Space staff, who were also familiar with the ongoing development of Gaslab.

However, our design direction faced some challenges during the early stages, as the absence of a building manager for Gaslab made it difficult for us to coordinate with those directly responsible for the building. Despite this, our team formulated a flexible and future-oriented renovation plan that aims to provide a strong connection between students and academic institutions.

Additionally, our proposal included methods for improving the indoor environmental quality in terms of lighting and temperature comfort. We also suggested strategies that can enhance energy efficiency. Moreover, we proposed a revised roof system which includes a system for water retention that can mitigate the accumulation of rainwater in the surrounding area and prolonged stress on the city's sewage system.

6.1.2 | Roadmap

The roadmap presents a clear overview of our process and achievements until the Midterm Demo (Figure 11). The image is a clear representation of our structured, yet iterative approach for the design. Before developing the initial renovation and repurposing proposal, our team conducted in-depth research on Gaslab's past and carried out interviews with former and current users of the building. These steps enabled us to better understand the characteristics, evolution and usage patterns of the building. Therefore, we were able to formulate a comprehensive program of requirements and set a direction for the repurposing project and future research.

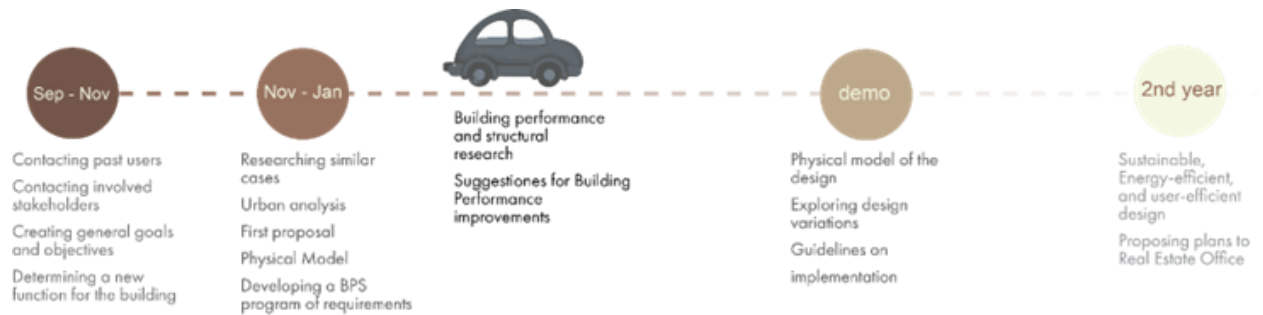


Figure 11

6.1.3 | Deliverables

Ground floor – Plan

The plan for the ground floor proposes an open space dedicated to three student teams. This space is envisioned as an **active and dynamic** environment that fosters collaboration between students, teachers and researchers. On the open, eastern side of Gaslab, flexible furniture strategies are implemented: **modular elements** that allow the layout to be easily rearranged to suit the teams' specific needs and preferences.

In the back room of the building, where the old boiler and the testing rooms of the former chemistry laboratory used to be, we proposed a small workshop and light assembly area. This addition supports a more **efficient workflow** for student teams as they are able to conduct hands-on activities within Gaslab itself. Therefore, the need to commute to the Innovation Space - which is currently not even connected to the building via footbridges - is minimized.

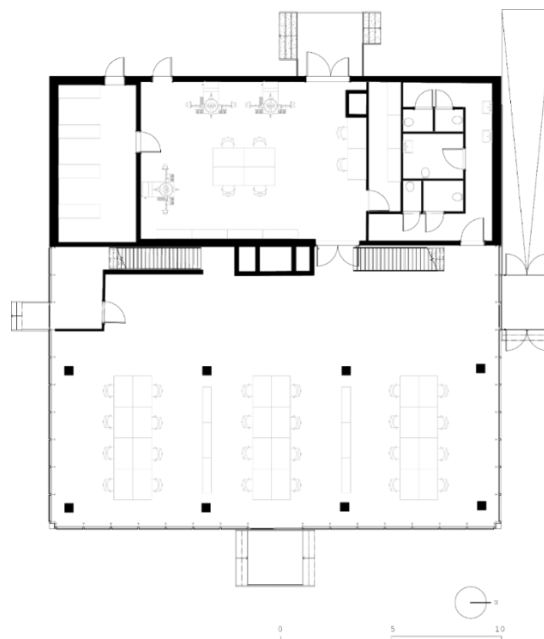


Figure 12

First floor – Plan

Similarly to the current use of the building, the open balcony area is envisioned as a lively space, where users can **relax and interact** during their breaks. This area also serves as an important threshold between the loud and lively downstairs workspace, that is focused on student team collaboration, and the more private and quiet upstairs room, which is designed for more private or focused meetings and presentations.

The western, closed room takes advantage of the existing thick division walls that isolate it from the rest of the building and provide the optimal conditions for its functions. Additionally, the space is conceived with flexibility and adaptability in mind. The large room can function as a venue for bigger meetings or private activities, but it can also be divided into three smaller spaces using the sliding walls installed along the back wall. The small and dark gas chambers situated along the sides are dedicated for storage.

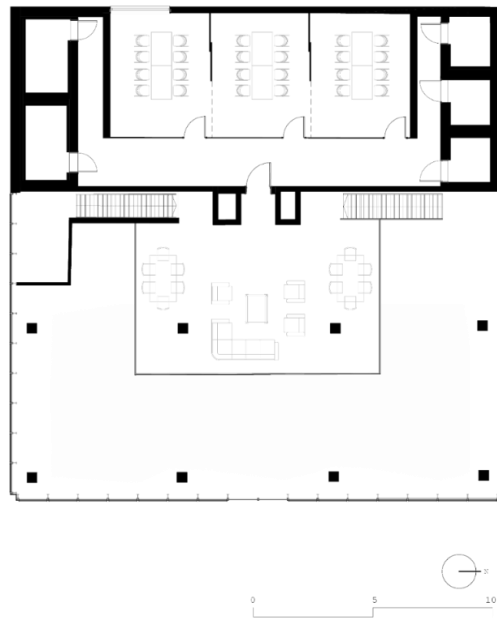


Figure 13

3D visualizations of the building

The exterior remains similar to the old Gaslab building, as preserving and showcasing the **original architectural feeling** of the campus buildings is a definitory aspect of our vision. The curtain wall, a key feature in the overall exterior look of the building, represents a major concern for energy losses in terms of heat loss through transmission. Therefore, a replica using a triple-glazed, energy-efficient

system that respects the same rhythm and division of the glass panes, is proposed as a substitute for the current, old curtain wall façade (Figure 14).

Moreover, the render showcases the implementation of the green roof, with water retention features. This addition slightly alters the proportions of the upper part of the façade, as the roofing system requires additional height. Nevertheless, the overall composition of the revised roof structure remains balanced and proportional in relation to the rest of the building elements, as it blends in through an extended roof band.



Figure 14

The interior design preserves key historical elements by protecting the original pipes and remnants still present within the building. Additionally, elements painted in the **representative colors**, such as the **iconic staircase and railings** are conserved and integrated into the more modern redesign of the building (Figure 15). The historically significant elements in vibrant colors are complemented by contemporary furniture and materials (Figure 16). This creates a strong visual communication and a soft transition towards the innovative feel that is introduced through the new interior design and organization.



Figure 15



Figure 16

Water retention roof system

Regarding improvements to the building envelope, the water retention roof system is an idea that was mentioned in a discussion with the Green Office (Figure 17). Their aim is to ensure that the buildings on campus function in a sustainable way. This Nature Based Solution provides an added aesthetic feature for the building, as it can be seen both from eye level and from inside the other, taller buildings. Additionally, this brings back some greenery in a space that is predominantly paved and functions as a parking lot, which is also an idea stated in the proposals for the green/blue structures of the TU/e.

Moreover, besides the environmental and aesthetic benefits, this roof system provides many functional advantages (FOAMGLAS®, n.d.). Therefore, during extensive periods of rain the water can be collected inside of the structure (Resilio, n.d.) of the roof and released later by controlled drainage. This reduces runoff, delays peak flow, and helps manage stormwater directly at the source (Nophadrain, n.d.).

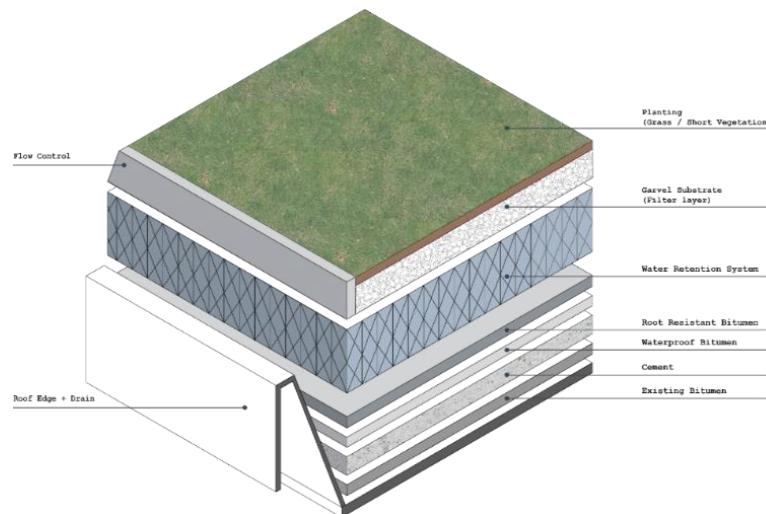


Figure 17

Idea for transforming Gaslab into a Living Lab

The discussion with the Green Office generated another idea for us: transforming Gaslab into a Living Lab, like the renovation of Atlas. As one of the most sustainable educational buildings in the world and the holder of a BREEAM Certificate (TU/e, n.d.), Atlas provided valuable insights in the process of **sustainable renovation** and implementation of innovative systems, such as lighting and climate systems that contribute to scientific research in the middle of a functioning office space.

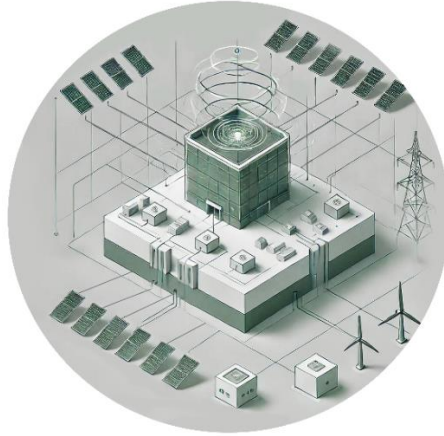


Figure 18

The idea for Gaslab was to implement an energy system that will efficiently generate power for the building to function in a sustainable manner. Our team considered implementing solar panels, but the urban analysis regarding the building heights showcased how this solution might not be viable due to the building being surrounded by multiple taller structures. Additionally, the integration of solar facade panels and energy recovery systems were also considered as alternative solutions to enhance the building's sustainability and as **opportunities for long-term research** and monitoring. By incorporating these technologies into the Living Lab evolution, Gaslab could serve as a testbed for developing strategies and frameworks for future, sustainable office building renovations.

Poster

The technical drawings and visualizations produced were compiled in a poster, which represented our main visual communication medium for the Midterm Demo. The aim of the poster is to convey a cohesive narrative of our visions and design input for Gaslab, making sure to highlight the rationale and process behind the chosen approaches and direction of the project.



The aim of this honors project is to give the old and historical Gaslab building a new look and feel while meeting the future needs of the university, particularly the needs of Innovation Space, which aims to have an "experience center" and provide new housing options for student teams. This will be done while maintaining the historical heritage and monumental elements of the building.

Our goal is to conduct a thorough analysis to develop a design that addresses and adapts to the current problems and needs of both students and working professionals. In this way, the renovated building will be transformed into a multifunctional space-flexible and easily adaptable to changes over time-while maintaining a clear and fixed purpose to ensure optimal functionality.

Additionally, it is this project's goal to improve the building's energy performance, structural integrity, and connection to the campus achieving a sustainable building with a meaningful presence on the TU/e campus.

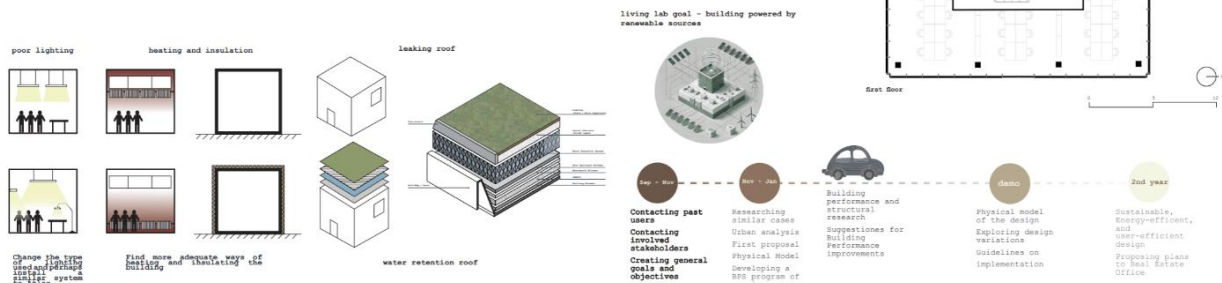
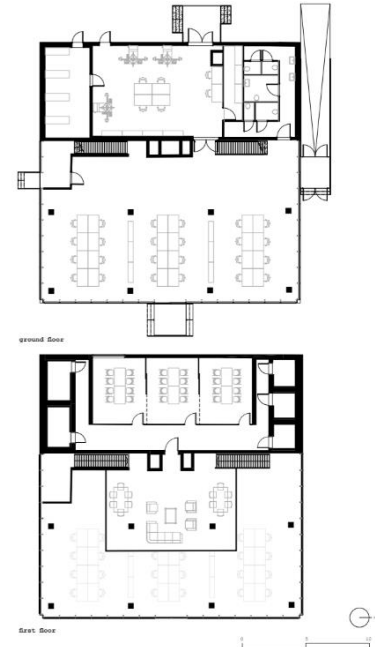




Figure 20

6.1.4 | Learning curve: Reflections and adjustments post-midterm

The midterm presentation marked an important milestone in our progress. This moment was also a significant turning point when our team realized that we were restricting ourselves in terms of Gaslab's future function. This had resulted from the uncertainties we had in our early stages and the difficulty in communicating with the right stakeholders. Consequently, our initial design was reconsidered in our next stages, in order to fully align with our long-term ambitions of creating something unique and innovative.

This realization led us to take a step back and reassess our direction. We opened discussions about integrating and combining multiple functions within the building, while seeking to reinforce our core concepts of flexibility and adaptability.

Based on the feedback we received during the Midterm Demo, our team also made the decision to **narrow our research focus** and redefine our objectives for the remainder of the first Honors year. Moving forward, we will focus on integrating new functional possibilities and developing adaptive design strategies that align more closely with our vision. This enables us to also explore ways to create a space that incorporates a multitude of disciplines, while making the most out of the restricted space within the building.

6.2 | Final Demo

6.2.1 | Introduction

When determining the name of the project we decided on NexUS as we want Gaslab to eventually become a center of innovation and collaboration, that connects the university to the exterior world. The location of Gaslab on the south side of campus, close to the Eindhoven city center, allows for a physical connection, reinforcing our idea of Gaslab becoming a bridge (Figure 21). Moreover, the history of the building, and its place and relevance within the history of the university, further reiterates the fact that Gaslab is a central meeting point, **connecting the past, present and future** of TU/e. By designing Gaslab as a meeting space **between art and technology**, we hope to bring together both worlds to inspire the building's users in their endeavors.

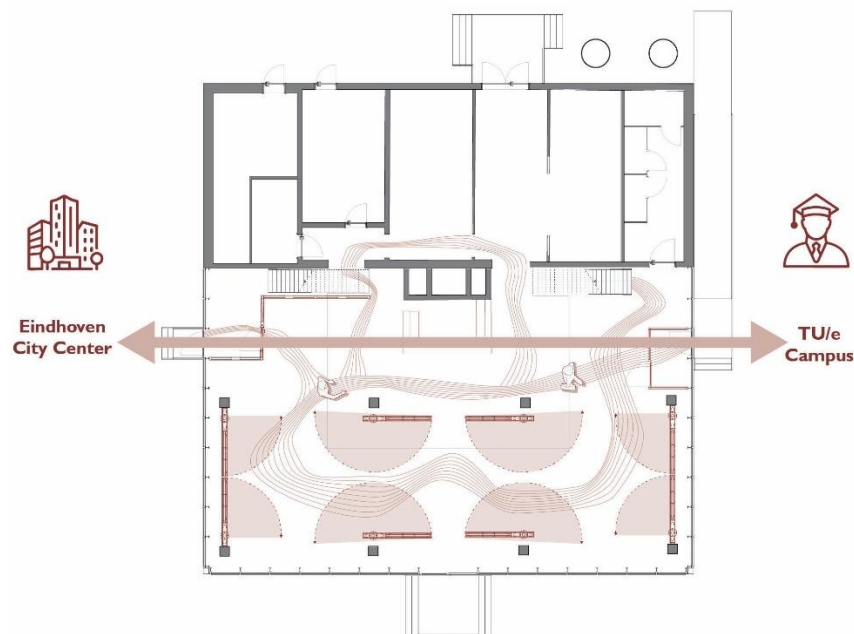


Figure 21

After the multiple directional changes faced in the time leading up to the Midterm Demo, our project took a major turn after an inspiring meeting with the TU/e Art Collection Committee. This is when the first vision of Gaslab as an art depot, that could showcase the vast cultural heritage of our university, was born. The spatial flexibility that lays as the foundation of a depot concept falls well within the existing goals of Innovation Space to transform Gaslab into a collaboration center on the TU/e campus. And as of lately, we initiated a line of collaboration with the Digital Twin Lab, with the aim of further fostering innovation in Gaslab, by planning dedicated spaces to the display of cutting-edge technology, such as VR equipment and hologram tables.

Our final design builds upon the core ideas of flexibility, multifunctionality and connectivity that were laid during the Midterm Demo. This goal is reflected in the distribution of the space and the materials

used in the renovation. With NexUS, we aim to shape not just a building, but a shared vision for what the **future of interdisciplinary collaboration** at TU/e can look like, thus inspiring current and prospective students and fostering a synergy with external institutes and companies.

6.2.2 | Inspiration

As the direction of the project settled towards creating an art depot within Gaslab, our inspiration took flight, as we started delving into case studies of the design concept we wanted to pursue. Surely, the most relevant examples that we set for ourselves were the Depot Boijmans Van Beuningen (Image 1 and 2) and Collection Pavilion in the Middleheim Museum (Image 3).



Image 1

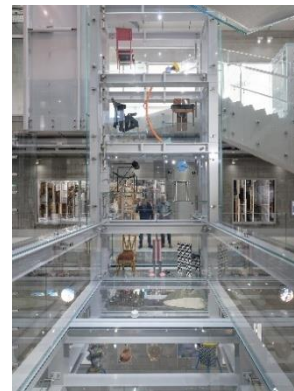


Image 2



Image 3

The most interesting feature that can be noted from these exemplar projects is how the design is able to reveal all these previously invisible artefacts in a unique and iconic **reflective building**: the visitor gets a glimpse of the vastity of the art collection, by passing through an everchanging selection of artifacts protected in **a multilayered and playful** glass display structure.

In order to apply the concept of adaptability and modularity to the idea of a depot, we took inspiration from the display panels installed by OMA / AMO in their new flexible exhibition space for the permanent collection of the Amsterdam Stedelijk Museum: the free-standing steel partitions create a **flowing pathway** for viewing art, thanks to an interlocking system used for the panels' mechanism.

This would encourage flexible circulation over the open floor area and foster enhanced lightness of the surrounding space.

6.2.3 | Deliverables

Ground floor – Plan

The ground floor was designed with movable panels (see section below), which allow the space to be modifiable, as can be observed in Figure 22 and 23. These panels also allow for more wall surface area in which to hang displays. By employing different combinations with the movable panels, the space can be easily adapted for many different functions. To support this flexible modularity an easily accessible storage space has been designed where support furniture can be stored, making the process of changing the space faster and easier. The storage on the left also communicates with the exterior which improves maneuverability. Various new spaces have been added including the hologram table room and the VR area – which can also be moved if needed. A reception desk has also been added to the design as it is Innovation Space's intention that there is a person that works in Gaslab and acts both as security and as a tour guide or support staff - there to answer questions and guide people through the Gaslab experience. The empty space between the storage space on the right and the hologram table room is to be kept empty as it is a part of the safety prevention measures of the building.

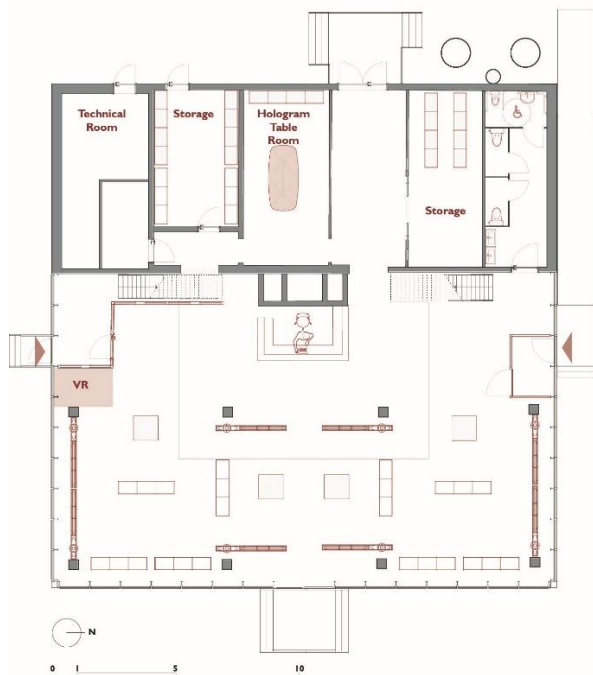


Figure 22

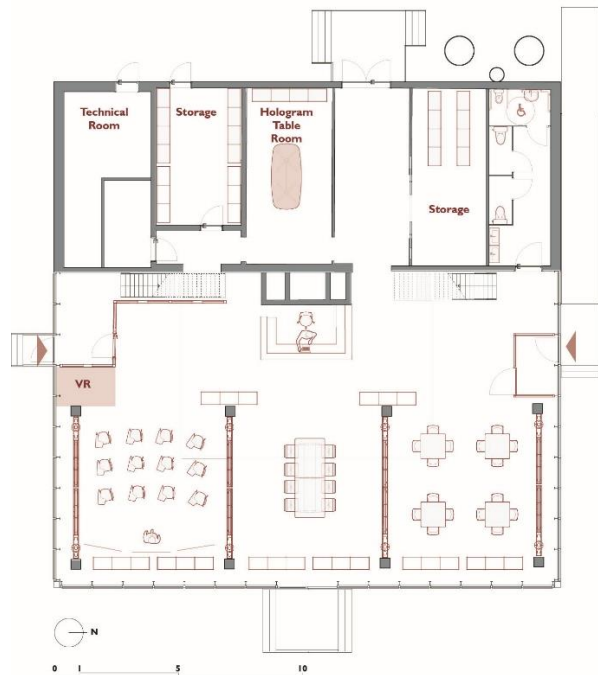


Figure 23

First floor – Plan

The first floor is arranged as more of a working area, with its own storage and a kitchen facility. However, to maintain the flexibility of the space all pieces of furniture used will be dividable, stackable and easily arranged into different combinations, maximising the utility of the space. Individual work pods have also been added in the far-right corner to allow for individual calls and meetings without disturbing the rest of the people working there. In the terrace it can be seen that a series of walls have been arranged, reducing the lounge area, in a U-shape. This design choice has been utilised to continue and maximise the exhibition space on the first floor. This floor also has roof access for maintenance purposes in one of the separate rooms located in the corners as can be seen in Figure 24.

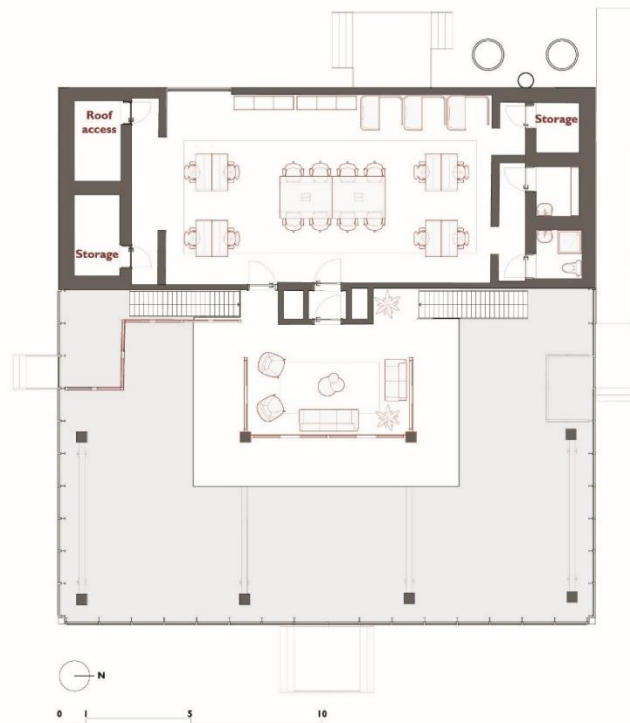


Figure 24

Elevations

The external envelope remains unaltered to preserve the historical character and value of Gaslab. One of the design's goals is to convey the **juxtaposition between old and new**, in a seamless transition that does not disturb the original look and feel of the space. The different façades of the building showcase the interplay between the fragility of glass and the dark masonry, representative of the hard industrial identity of Gaslab as a former chemical laboratory (Figure 25). The extensive glazed surfaces allow the external observer to glance at the dynamic workflow and everchanging exhibition that will be housed in Gaslab, so we wanted to take advantage of this structural feature of

the building to showcase the adaptability and versatility of the new space. Moreover, the transparency of glass will symbolize **temporal continuity** between Gaslab's various identities throughout time, giving the visitors the ability to see through to the past, while also catching a glimpse of future innovation potential.



Figure 25

A particular feature of the final design consists in switching the solid wall, which is currently installed around the left red staircase for fire safety purposes, with a fully glazed wall to bring back the original symmetry of the two staircases that lead to the first floor (Figure 26). The idea originated from our initial desire to restore this distinctive trait of the building's style, also after learning that the symmetrical look was a largely appreciated feature among Gaslabs past users, but the fire safety regulations posed a severe limitation to our design freedom. The choice of installing a glass partition is our proposed solution to merge an aesthetical attribute with functional needs.

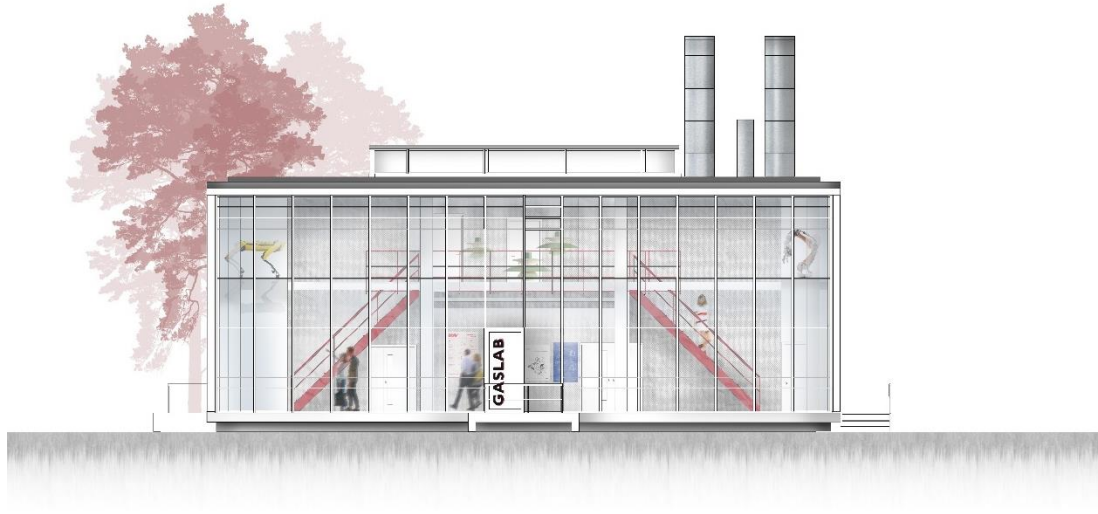


Figure 26

Sections

We envision the future Gaslab to be an **intersection point between different disciplines** and personalities from all different backgrounds. This entails the creation of a highly adaptable space that actively responds to its multifunctional needs. Gaslab will foster a blend of different atmospheres, by housing a combination of different environments, such as meeting areas and experience centers (i.e. hologram table) on the ground floor, as well as working spaces and lounge areas on the first floor. The distribution of the function zones across the building are also dictated by the existing structural layout of the building and also by variable factors, such as daylight availability (Figure 27).



Figure 27

Additionally, we want to take advantage of the massive space offered by the double height ceiling on the ground floor, by creating extensions of glass boxes that enclose the two lateral entrances and enhancing the **verticality** of the interior. The installed structures will serve to display more artworks,

such as paintings and hanging objects (Figure 28). Our goal with this design implementation is to stimulate the visitor's awe right from the entrance and immediately transport them into a suggestive experience within a building that is not only **multifunctional** but also **multidimensional**.



Figure 28

Wall system

One of our main stakeholders – Innovation Space – requires flexibility in the space and the ability to adapt it to different functions. A second important stakeholder – the Art Commission of TU/e – seeks a space to store the university's art collection. Both stakeholders align in their priorities, emphasizing the creation of an exhibition space that honors the university's history.

To accommodate these functions, we focused on designing flexible solutions for space division and art collection storage. The design features a steel frame with glass covering and two movable panels inside, allowing for additional storage and space division options (Figure 29). Additionally, the panels can rotate 180 degrees to create varied spatial configurations and meet the needs of different users.

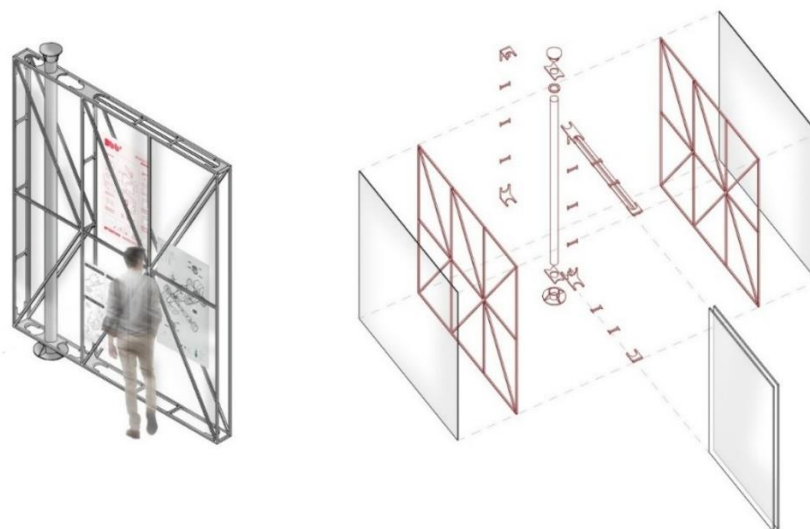


Figure 29

Moreover, the design of these partition walls allows for the easy attachment of shelves for larger artifacts and robots. The system is anchored both to the ground and to an additional beam that connects to the structural columns of Gaslab. Nevertheless, the original character and feel of the former high-pressure laboratory is preserved through the industrial look of the movable panels.

Additionally, inspiration for this rotating structure was taken from an analysed project: Mumokuteki Bookstore. This system was especially chosen for its non-invasive structure that does not imply major changes to the existing building, providing the opportunity for removal without altering the space. Additionally, its smart and minimalist approach to spatial division aligns with our intention to create a flexible, reversible intervention that respects the original architectural identity of Gaslab while fulfilling contemporary functional needs.

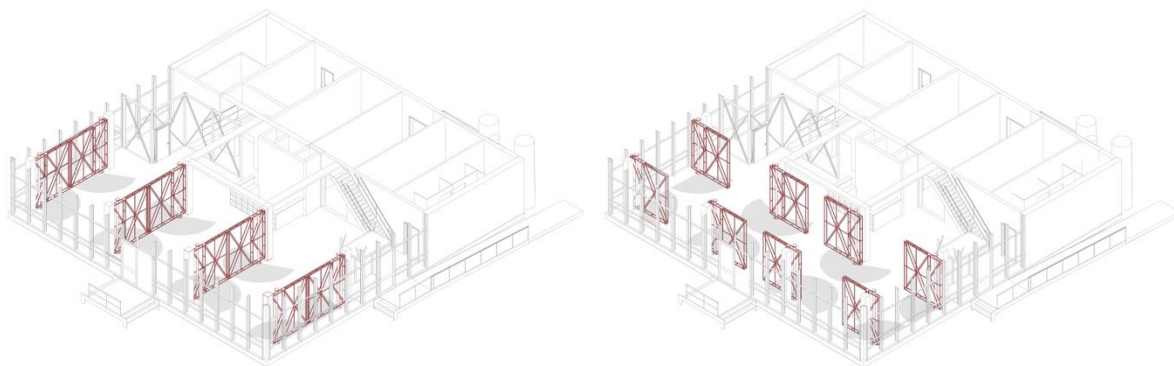


Figure 30

This system can be used in many configurations, two of them being showcased above, in *Figure 30*. The flexible wall design is envisioned to adapt to different scenarios through the 180 degrees range of rotation. Therefore, the first axonometric illustrates the interior space of Gaslab divided in 3 partially enclosed spaces, by keeping the panels in straight position, aligned with the columns and the span of the beams. This division facilitates the inclusion of different workspaces with flexible, foldable furniture, and provides a sense of privacy for the undergoing activities. Hence, this interior configuration of Gaslab can both support the exhibition and foster workshops or other presentations and interactive activities.

From the first presented position, the walls can be rotated 90 degrees to get a more open space, as illustrated in the second axonometric view. This new space allows for circular movement and a free exploration flow between the exhibits. Additionally, the space can gather larger artefacts or display other elements on pedestals/supports placed in the open area.

Nevertheless, this flexible wall system allows also for more playful and dynamic configurations, based on the activities happening inside. Therefore, the panels can be placed in intermediary positions which create playfulness throughout the floorplan by projecting dynamic diagonals into the space.

Renders

The proposed design preserves the exterior look of the building and aims to revitalize Gaslab by curating the surrounding area, through the addition of greenery and the reduction of parking spaces, to reduce car-centered transport and promote green modes of transport (Figure 31). This will make the building more appealing to visitors and more comfortable for its regular users.



Figure 31

The aim of the interior design is to create a visually striking environment that valorizes the displayed artifacts from the TU/e Art Collection. The distinctive industrial character is conserved, thus enhancing the cultural significance of Gaslab, and the contrast of the concrete surfaces with the representative red staircases and railings is highlighted with the addition of new furniture that recalls the same vibrant color, such as the reception desk in Figure 32. The implemented glass panels function as dividers but still allow a certain level of connection between adjacent zones of the floor area, as well as maintaining visual communication with the outdoor environment, allowing natural daylight to permeate the room (Figure 33).



Figure 32

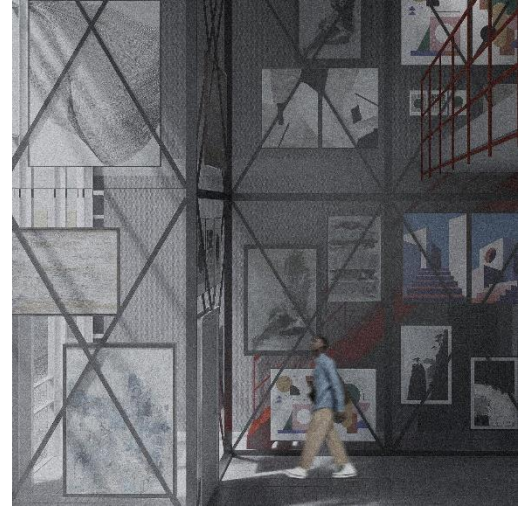


Figure 33

The balcony on the first floor is partly enclosed with glazed panels to isolate the space from the possible noise coming from the ground floor and create a comfortable lounge area, as well as a small-scale gallery, by exhibiting art pieces on the external sides (Figure 34). We envision this space as a hub for social interaction, as well as a pleasant meeting area for external collaborators.



Figure 34

Website

As part of our renovation project for the Gaslab building, we decided to create a website to serve both as a digital portfolio and a dynamic project diary. This platform allows us to clearly present our goals, document our progress, and showcase our results in an accessible and engaging format.

The website begins with a homepage that introduces the overall premises of the project and highlights key features of the Gaslab building. This section serves as a reference point for visitors, establishing the context and significance of our starting position.

To document our progress, we have structured the website to include dedicated pages for each major delivery and project milestone. At this stage, we have created specific sections for the mid-term and final demos of this year. By organizing the content this way, we ensure that the evolution of our work can be followed step-by-step in a coherent and transparent manner.

In addition, the website includes a page focused on the building and urbanistic analysis we conducted on the Gaslab. This section presents our findings and methods, including the measurements collected for the Building Physics analysis, providing insight into the technical foundation of our renovation approach.

We have also devoted a section of the site to the historical background of the Gaslab, as described in Section 3.4. This part features a detailed timeline, archival photos, and explanations of the various periods in the building's past. Our intention is to highlight Gaslab's cultural and architectural importance, portraying it as a symbolic time capsule within the university's heritage.

A team page introduces the members involved in the project, along with current collaborators. This includes the various institutes, faculties, and committees we are working with, as well as a description of their specific roles and contributions to the renovation process.

Finally, the website features a contact form, enabling anyone interested in the project—whether they have questions, feedback, or would like to get involved—to easily reach out to us. This helps facilitate direct communication and fosters a sense of openness and collaboration.

Ultimately, we envision the website as a central hub that gives our project visibility, helps us engage with the wider community, and provides our collaborators and potential future investors with a clear overview of what we have accomplished, what we are currently working on, and our plans for the future. We believe that this digital platform is not only a useful tool for documentation and outreach but also a creative and interactive way to celebrate and share the story of our work.

To visit the website, click the following link: [Gaslab NexUS](#)

6.2.4 | Future steps

Indoor Environmental Quality

Based on the findings from the BPS measurements, we started brainstorming design solutions targeted at improving the indoor environmental quality. Some of the prompts include:

- Installing controllable blinds for lighting regulation, which would also serve when a dark room is needed for on-screen presentations;
- Implementation of acoustic buffering surfaces, for instance inside the partition panels on the ground floor. Transparent acoustic panels can be explored as a solution to keep the visual characteristic of our new wall system;
- Replica of the façade with a curtain wall for additional insulation, or addition of a secondary glazing layer, but it would add undesired thickness to the building envelope and cause visual obstruction.

We will initiate more collaboration with retrofitting experts, as well as curators of historical buildings, to properly tackle the interior quality standards of Gaslab and improve the comfort levels of the users.

VR experience

One of our key collaborators in this project is the Digital Twin Lab from the EAISI Institute. Together, we aim to explore the integration of Virtual Reality (VR) technologies into our renovation project, with the goal of creating a unique, immersive experience that enhances both the functionality and outreach of the Gaslab space.

Several potential applications for VR within the project have been discussed. One of the most exciting possibilities involves virtually expanding the Gaslab building to create additional exhibition space. This virtual extension would allow for the display of a larger portion of the TU/e art collection, or works from various faculties, without being constrained by the physical limitations of the building. Furthermore, VR would enable the exhibition of delicate or sensitive pieces that might otherwise be at risk if exposed to environmental conditions, thereby broadening the range of items that can be presented to the public.

Another concept we have explored is the creation of a virtual model of the Gaslab and the broader TU/e campus. This model would offer visitors the opportunity to explore the campus and its architecture across different historical periods, showcasing the various stages of its development. By doing so, it would allow users to experience the heritage and evolution of the university from the unique vantage point of the Gaslab, thereby deepening their connection with the space and the institution.

Additionally, we have considered the use of VR to create a functional digital twin of the Gaslab. This tool would allow us to simulate renovations and reconfigurations of the building in real time, making it easier to visualize changes and optimize the design. Since the long-term goal for the Gaslab is to become a flexible, multifunctional space, a VR model would be instrumental in testing different layouts and spatial arrangements for various types of events. It would also serve as a compelling visual tool for communicating our vision to stakeholders, partners, and potential investors.

Among the ideas discussed, the most feasible and immediate project is the development of a virtual campus model with a historical focus. This initiative aligns with an ongoing plan already envisioned by the Digital Twin Lab and has therefore become our primary objective. Given the scope and complexity of modeling the entire campus through its various eras, we have collectively decided to start by focusing on a more manageable section. Specifically, the virtual model will feature the Gaslab and its surrounding area, changing over time to reflect the history of the campus, the rest of the campus will be modelled as it currently is in the present. This approach allows us to create a meaningful and educational experience while maintaining a realistic workload.

We believe this virtual campus project offers significant added value—not only to our own renovation initiative but also to the university as a whole. It presents a compelling way to engage with TU/e's

history, while also providing practical benefits. For instance, prospective international students could use the virtual tour to explore the campus remotely, and individuals with mobility challenges would gain improved access to its spaces.

To realize this project, both our team and the Digital Twin Lab will collaborate on building the virtual model. We are also exploring the possibility of involving students in its development, whether through dedicated courses, student teams, or as paid assistants. Looking further ahead, we hope to eventually expand the project to cover the entire TU/e campus through its different eras, creating a rich, interactive archive of its architectural evolution.

While our current focus is on the historical virtual campus, we remain enthusiastic about pursuing our other VR concepts in the future, such as the virtual expansion of Gaslab for larger-scale art exhibitions. We believe that these digital innovations offer a powerful way to extend the physical and cultural reach of our project, reinforcing its relevance both within the university and to the broader community.

6.2.5 | Learning curve: Reflection towards Final Demo

With the Final Demo and the submission of all our deliverables, we have achieved another important milestone for our repurposing project of the former high-pressure laboratory. Throughout the year, we embarked on a continuous process of learning, testing out different solutions and responsibly assessing our progress both on an academic and personal level. This iterative process had a fundamental role in continuously pushing our design and creativity in order to fulfill our goals in creating a modular, flexible, but also innovative and user-centered space. It was also a great opportunity for stepping outside of our comfort zone and exploring new possibilities in order to deepen our understanding about the building and learn to accommodate and work ‘around’ an existing structure.

Despite many challenges and obstacles that we encountered along the way, our efforts led to a future-proof design that aligns not only with the wishes of three stakeholders – Innovation Space, TU/e Art Collection and the Digital Twin Lab - but also with our vision of integrating a new, innovative and future resilient function, while preserving and highlighting the look and feel of the heritage building.

Additionally, the inspiring interactions in the time following the Midterm Demo, with representatives from the TU/e Art Collection and Digital Twin Lab, led to a fundamental decision in our process: setting the function and a clear direction for the repurpose. These collaborations enabled us to confidently draft the plans and were a key entity in shaping and validating our ideas.

The final design represents an important intersection point between the different stakeholders we involved and our own vision for the project. We are now actively collaborating with stakeholders to bring both history and innovation together in Gaslab. By combining art and technology, the new exhibition space that will be housed in Gaslab effectively increases the visibility of the Art Heritage Collection and department, exposing their work and the artifacts that the university possesses to a wider public.

Increasing the awareness and accessibility of the building also enhances the exposure of students and the residents of Eindhoven to the rich history of the university, perhaps even attracting new public, including potential investors and collaborators. Our aim is to attract the interest of the current building managerial department, including Innovation Space and TU/e Real Estate, to promote and initiate a concrete renovation plan for Gaslab, as there is currently no set date for its departure. We believe that by revamping and reactivating the building's functionality and improving the quality of its surrounding areas, Gaslab will become the new bridging point between external visitors and the university's community, virtually connecting the vitality of our campus with the dynamic cityscape of Eindhoven.

Over the past year, we worked on the plans, spatial organization, and stakeholder needs to ensure that the future design will fulfill one of our main goals for the building: creating a space that fosters interaction, collaboration, and cooperation. Ensuring the space could be used in flexible and diverse ways was a priority, but creating a pleasant and enjoyable workplace was also an important consideration. Both aspects contributed to shaping a space where people can meet and collaborate.

The spatial organization was primarily guided by user needs and comfort. At the beginning of the process, we conducted several interviews with both current and former users of the building. Based on the insights gathered, we developed a new layout and design aimed at making the space comfortable and functional for everyone.

This honors year was an important starting point for our project. We learned a great deal, not only about the building itself but also about collaboration with stakeholders, teamwork, and working on real-life projects. This knowledge will be valuable for future phases of the project, where we aim to focus more on sustainability and circularity. We also hope to open the door to further consultations, particularly with renovation experts, which would help make the project more feasible.

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3. <https://middelheimmuseum.be/en/content/collectiepaviljoen>

9. | AI Usage Statement

Parts of this report were reviewed using an AI tool - ChatGPT - to check grammar and structure and ensure that the content is clearly organized. All ideas, experiences, research, drawings and project content are based on our personal work and contribution.

AI was not used to generate content or research. This AI tool was also sometimes used during our brainstorming sessions, to support and steer idea generation, but all output was critically assessed and then checked with other resources to ensure validity of content.

Moreover, *Figure 18* was generated with the help of AI and then adapted and refined with Adobe Photoshop. This technique was only used for this specific figure, as it was a concept without a clear visual representation and the AI tool helped in portraying the diagram.

10. | Appendix: Individual contribution

Name: Maria Barnechea

Student nr.: 1932330

At the beginning of the project, it was initially challenging for me to find a clear role within the team due to the specific and multifaceted nature of the work, especially in the field of architecture. However, over time, as the project expanded and developed, I became actively involved in several key areas.

One of my primary responsibilities has been the development and content curation of our project website. After deciding the content of the website as a team I was in charge of structuring the content, deciding what information to include, and how best to communicate our work in a clear and engaging way. As a result, I took on the task of compiling and editing the text, images and other media, either previously created by my peers or generated specifically for the website, for various the sections, such as the homepage, historical overview of the Gaslab, and pages dedicated to specific deliverables. I was also tasked with all the technical and design aspects of the website.

In terms of team process, I took an active role in external communications following the initial outreach phase, which was performed by the whole team. I handled incoming emails, followed up with stakeholders, and ensured timely and professional responses. I also participated with my team members in several meetings with different interested or relevant parties to help build relationships and explore opportunities for support or involvement in the project.

Additionally, I was actively involved in the collaboration with the Digital Twin Lab, where I contributed to brainstorming and developing concepts for integrating VR into the project. I helped research comparable projects and gathered visual and technical references to support our discussions and proposal development.

Another area that I handled was contributing to the research on the history of Gaslab was. From the sources that were provided to us and the relevant resources online I gathered historical information, organized it into a coherent narrative, and generated a written explanation. Under the guidance of Valentina, I also assisted in taking physical measurements of the building for temperature, light and acoustics, supporting the team's efforts in analyzing the building's metrics.

Finally, I contributed to the writing of reports and throughout the project, I participated in group discussions and decision-making sessions, where I offered input on the overall direction of the project, helping the design team in guiding and defining ideas to complete and visualize the project

Name: Andreea Vasile

Student nr.: 1988247

Throughout the project, I naturally took on the designer role by generating design iterations focused on layout strategies and creating visual communications accessible to both experts and non-experts. I also used my enthusiastic and ambitious personality to create an open and inclusive atmosphere, where everyone's strengths were highlighted and encouraged. In parallel, I often took on managerial tasks to help structure and streamline the team's workflow. Throughout the year, I also contributed to the oral presentations, only becoming a more defined role towards the end, when I was presenting and creating a script with Valentina.

I actively supported the team process by taking responsibility for documenting meetings and contributing to team planning. I regularly took minutes and provided an overview of discussions and upcoming steps. I also contributed to breaking down complex tasks into smaller, achievable parts, that were translated into action points for the team. In the beginning, I participated in organizing external meetings and interviews with stakeholders, by reaching out to them and preparing question frameworks that allowed for a structured and easy interaction.

One of my biggest contributions was conducting research and experimenting with layout and design strategies for planning a successful exhibition. I gathered and synthesised inspiring projects and relevant case studies. I had an active participation in our brainstorming sessions and translated my findings into concrete ideas that aimed for flexibility, adaptability and innovative display systems. Therefore, our team was able to find a suitable and future-proof solution for Gaslab, that aligned with both the stakeholder's requirements and our vision.

I also contributed towards team deliverables by creating detailed drawings and visualizations throughout the process. I experimented with software like Revit, Archicad, Twinmotion, Rayon and Adobe Creative Cloud, to deliver technical drawings such as floorplans, elevations, explanatory diagrams and renders, that effectively communicate our ideas to the public. I helped in preparing a file for laser-cutting and participated in the assembly of the physical model. I contributed extensively to the Midterm and Final Demo posters.

Additionally, I researched and then reconstructed the historical evolution of Gaslab through a visual timeline. I studied information about Gaslab, as well as frameworks and strategies regarding the campus layout. I conducted, along with Marta, an urban and spatial analysis, which I later reflected on and documented in the final report. Therefore, I proposed architectural principles that guided us through our design process aiming to preserve the character of Gaslab while introducing its new functions and ensuring the design remained responsive to the site's spatial conditions.

Finally, my work also focused on sustainable and innovative design practices. I regularly stressed the importance of implementing technical aspects in the repurposing project. I researched concepts for a water retention roof and proposed solutions for improving environmental performance, including renewable energy generation.

Name: Marta Zawadka
Student nr.: 1966979

During the project, I fell into several different roles. One of the main ones was being part of the design team, which was supported by my architectural background. Throughout the past year, I contributed to brainstorming sessions and creating visuals to better showcase the main points of the project. Additionally, even though we didn't specify group roles, I was mostly taking on the tasks of a leader supporting group organization, planning, and creating an inclusive environment. At the beginning of the project, I also contributed by contacting stakeholders and conducting research.

Due to my studies, competences, and personal preferences, I became part of the design team. Our goal was to create a new functional plan for the Gaslab. Together with the team, we focused on identifying the strengths and weaknesses of the building, then addressed existing challenges with long-lasting solutions, fulfilling the needs of different stakeholders. I took an active part in designing the spatial organization of the Gaslab, searching for different solutions and options. In the final steps, I also contributed to brainstorming ideas for panel options that could be implemented in the building.

Another task I contributed to was the creation of the digital model and deliverables for both the midterm and final demo, as well as track meetings. In collaboration with Andreea, we created the urban analysis that helped identify starting points for the design. I also helped create AutoCAD files for laser cutting and later assembled the physical model. For the midterm demo, I worked on rendering, creating floorplans, and putting everything on the poster, while ensuring clarity and understanding of the project. I worked on the digital modelling of the new designs in Rhino showcasing the final design and testing different variants. I visualized the final design in detail and created several renders to show different impressions, present our future plans, and enhance the quality of the deliverables. I also worked on sections and diagrams. Throughout the year, I created a visualized timeline that showed our progress and explored different software, including AutoCAD, Revit, Rhino, Adobe Cloud, Twinmotion, and D5.

The second important task I took on was administrative and planning responsibilities. I kept track of meetings, scheduled regular weekly team meetings, and created agendas for most of them. I encouraged brainstorming at key project milestones and aimed to create an inclusive environment so everyone could contribute using their strengths, background knowledge, and ideas. I tried to leave space for others to discuss, implement changes, and share ideas. In the first months, I supported stakeholder meetings by providing the team with a table listing meeting dates and outcomes.

At the beginning of the project, I also conducted research on TU/e development plans and the history of the Gaslab. I contributed by contacting involved parties and presenting during track meetings and the midterm demo. Later, as the team shifted tasks to focus on individual competencies, I presented and contacted external parties less, allowing others to take over while I focused more on developing my own.

Name: Valentina Wu
Student nr.: 1949292

Initially, we did not establish specific roles, as we were trying to find a direction for our project and figuring out our goals as a team. At that time, I contributed to contacting stakeholders and participated in meetings with them, along with the rest of the team, to better understand the needs, constraints, and opportunities surrounding Gaslab. I also contributed to analyse the findings from these interviews and elaborate them in a cohesive way, which helped creating a foundation for more targeted design requirements later on.

As the project developed, I gradually found my place by naturally falling more into the technical aspects of the project, particularly within the building physics field. I applied my specific study background and knowledge, by creating a detailed measurement plan for the Building Physics Lab. I outlined the equipment, methodology, and objectives for the assesement of the indoor environmental quality of the building, with a focus on lighting, temperature, and acoustics. Following this plan, I took the leading role during the on-site measurement we conducted in Gaslab to assess the indoor environmental quality of the building. I was also responsible for analyzing these findings by checking their compliance with a Program of Requirements, which I developed by researching recognized frameworks and standards. I made sure to extract relevant conclusions and considerations that could be easily translated into concrete design recommendations.

As an important part of my personal development, I contributed to presenting and engaging with questions during track meetings throughout the entire project, and in collaboration with Andreea, we took the responsibility of leading the presentations towards the end of the year. I contributed to the definition of goals and desired project outcomes, by actively participating in team brainstorming sessions and providing my input in decision-making. I prioritized building on others' input, making sure each of us could take on responsibilities that matched our individual strength. I believe this contributed to creating a respectful and motivated team atmosphere, which was essential to the overall success of the project.

Moreover, I often took the initiative on organizational tasks to manage files and information within our shared folder. For instance, I was able to streamline our correspondence, which was especially important for managing our broad matrix of stakeholders and maintaining contact with them, thus contributing to making our workflow more efficient.

In terms of deliverables, I contributed to creating graphic visualizations for the Midterm Demo, specifically the diagrams regarding future steps and outcomes for the BPS part. However, my biggest contribution towards the final output is structuring and writing large sections of the report. My goal was to translate our process into a clear and concise documentation. I reflected on the challenges we faced throughout the project and made an effort to convey the rationale behind key decisions that ended up shaping our final design proposal. My goal was to make the report not only a summary of outcomes but also a transparent overview of the collaborative journey we embraced as a team.